

Thulium

Tm (Tm I). Strongest classified transitions below 2 μm (NIST): 72 levels, 73 lines.

Tm⁺ ionisation limit 49880.57 cm⁻¹ = 6.18440 eV (200.479 nm)

Arrow label: vacuum wavelength | reduced dipole matrix element |*U*||*er*||*U*⟩ in ea₀
Matrix elements are measured or NIST values: * = high-accuracy theory, ≈ = model calculation.
Arrow width grows with the Einstein coefficient A. Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation, *n* = 70.
Level caption: energy (NIST, or a measurement of the last 20 years where one exists) and measured lifetime (where one exists).

